Zhongxin Hu

PhD Candidate at Eindhoven University of Technology

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Research Profile

My research focuses on sustainable operations and supply chain management. I study how firms can design circular business models, such as remanufacturing and refurbishment, as well as other operational strategies that create economic value while improving environmental and societal outcomes. My work primarily uses analytical modeling and optimization.
My broader research interests include pricing and equilibrium analysis in queueing systems and sequential games.

Academic Experience

Eindhoven University of Technology **Eindhoven, NL**
PhD Candidate in Operations Management *July 2023 – Expected 2027*

- Supervisors: Zümbül Atan and Christina Imdahl
- Group: Operations, Planning, Accounting & Control (OPAC)

INSEAD **Fontainebleau, FR**
Visiting Researcher *Sep. 2026 – Dec. 2026*

- Host Professors: Luk van Wassenhove
- Group: Sustainable Business Initiative & Humanitarian Research Group

Beijing University of Posts and Telecommunications **Beijing, CN**
M.Sc. in Mathematics *Sep. 2020 – Jun. 2023*

- Supervisor: Yongjiang Guo
- Thesis: *Information Disclosure and Reference Dependence in the Cooperative Newsvendor Problem*

Nanjing University of Posts and Telecommunications **Nanjing, CN**
B.Sc. in Applied Statistics *Sep. 2015 – Jun. 2019*

Working Papers

- **Profit Meets Purpose: Refurbish-to-Lease Model for Medical Essentials**
  Zhongxin Hu, Christina Imdahl, Yeqing Zhou (RSM), Zümbül Atan.
  Status: Revision in progress.
  Conference presentations:
  2026 INFORMS Annual Meeting, San Francisco, USA(upcoming); POMS SCM Research Webinar, Online; OPAC Conference, Heeze, NL; Circularity and Sustainability Workshop (Poster), Utrecht, NL.
  2025 Beta Symposium, Soesterberg, NL.

Publications

[1] **Zhongxin Hu**, Christina Imdahl, Zümbül Atan. Strategic Selection of Remanufacturing Business Models: A Consumer Perception Perspective. *European Journal of Operational Research.* Accepted for publication.

[2] Jian Cao, Yongjiang Guo, **Zhongxin Hu** (2023). The Effect of Loss Preference on Queueing with Information Disclosure Policy. *Methodology and Computing in Applied Probability* 25: 75.

[3] **Zhongxin Hu**, Yongjiang Guo (2023). Cooperating Strategy of Newsvendor with Advertisers Under Partial Quality Disclosure. *Journal of Systems Science and Mathematical Sciences* 43(3): 667–682.

Teaching Experience

Bachelor's Thesis Supervision

Eindhoven University of Technology

- Supervised 4 Bachelor's End Projects, 2024/25–2026/27.

Teaching Assistant

Eindhoven University of Technology

- Fundamentals of Operations Management, BSc, Q3 2024/25 & 2025/26.

Beijing University of Posts and Telecommunications

- Reinforcement Learning: Theory and Applications, MSc, Summer 2021.
- Probability Theory and Stochastic Processes, BSc, Spring 2021.
- Advanced Mathematics, BSc, Fall 2020.

Research & Project Experience

LINear to CIrcular Transition (LINCIT) **July 2023 – June 2026**

- Conducted research on operational decisions in circular business models, including remanufacturing and refurbishment.
- Presented research progress to academic and industry consortium partners.

Post-Graduate Innovation and Entrepreneurship Project **Apr. 2022 – Nov. 2022**

Project Leader

- Developed analytical models combining sequential game theory, prospect theory, and contract theory to study computational-resource allocation in blockchain systems.

Awards & Honors

- **Best Presentation Award**, OPAC Conference, Heeze, NL, 2026.
- **Outstanding Graduate Student Award**, Beijing University of Posts and Telecommunications, 2023.
- **Provincial First Prize**, China Contemporary Undergraduate Mathematical Contest in Modeling, Jiangsu Province, 2017.
- **Merit Undergraduate Student Award**, Nanjing University of Posts and Telecommunications, 2016.

Academic Service

- **Session Chair**, INFORMS Annual Meeting, Atlanta, USA, 2025.
- **Reviewer**, Journal of Systems Science and Mathematical Sciences, 2023.
- **Research Mentor**, Lumiere Education, 2026.

Last updated: October 2026